

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Operating Systems**

Code No: **CACS251** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is an operating system? Is "Multitasking" operating system better than "Batch" operating system? Justify your answer.
3. What is role of scheduler in operating system? Differentiate between short-term and long-term Scheduler.
4. What do you mean by Race condition? Consider two concurrently running process P1 with statement S1 and P2 with statement S2. Suppose we require that S2 be executed only after S1 has completed. For this synchronization need, write a semaphore solution.
5. What are CPU scheduling criteria's? Consider below table of four processes under multi-level queue scheduling. Queue number denotes queue of the process. Priority of queue 1 is greater than queue 2. Queue 1 use Round Robin with time quantum $t=2\text{ms}$ and queue 2 use FCFS. Draw Gantt chart to illustrate their execution and calculate average waiting and turnaround time also.
6. What is thrashing? Explain three memory management requirements?
7. Given five memory partition of 100KB, 500KB, 200KB, 300KB and 600KB (in order). How would the first fit, best fit and worst fit algorithm place processes of size 212KB, 417KB, 112KB, and 426KB (in order) in fixed-size partition? Which algorithm makes the most efficient use of memory?
8. When programmed I/O is suitable than other I/O handling techniques? Explain different type System Threat.

Group C

Attempt any TWO questions[2x10=20]

9. Consider the following System: Process Allocation Max Available [A, B, C, D] [A, B, C, D] [A, B, C, D] P0 0 1 10 0 1 21 5 2 0 P11 0 0 01 7 5 0 P21 3 5 42 3 5 6 P30 6 3 20 6 5 2 P40 0 1 40 6 5 6 Answer the following questions using Banker's algorithm: a) What will be the contents of matrix need? b) Draw the safe sequence? c) If the request from P1 Arrives for (0, 4, 2, 0) can the request be granted immediately?
10. How second chance page replacement algorithm differ from FIFO page replacement policy? Explain the concept of belay's anomaly with suitable example.

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- 11.** What is clock synchronization? Explain how physical clock is synchronized by Cristian algorithm and logical clock is synchronized by Lamport's algorithm with suitable example.